Both copies of BUB1B are broken. Only the second copy can still be helped — so the medicine must fit *that* copy.

Rank 1
Track 1 oracle confirmed `BUB1B` in *trans*

8
clinical HPO terms, incl. rhabdomyosarcoma & muscle atrophy

top 0.99%
mouse twin $Bub1b^{H/H}$ of 76,699 MGI genotypes (p = 0.026)

Copy #1 — `stop_gained`

- Premature stop codon, exon 17 of 23
- Canonical nonsense-mediated decay target
- Loses the whole C-terminal kinase domain

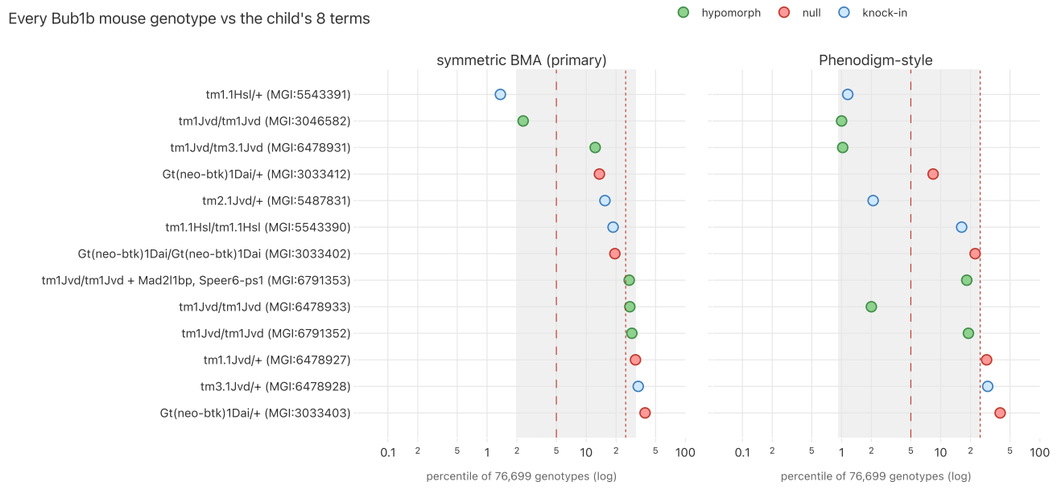
vs

Copy #2 — `missense_variant`

- One amino-acid swap in the pseudokinase domain
- Still makes BubR1 protein
- The only reason a biallelic-null-lethal gene left a living child

BubR1 is the cell's seatbelt inspector

It holds division until every chromosome is buckled in. With too little of it, daughter cells get the wrong chromosome count — mosaic variegated aneuploidy, premature senescence, muscle wasting, and an embryonal rhabdomyosarcoma.



The child's 8 symptoms, scored against every mouse genotype in MGI, select a hypomorph — not a knockout. report.md §3.2 (entry 150); figure step9-e2-rank

Every therapeutic decision for this child runs through Copy #2.

Caveat *Mechanism is read from models and a matched mouse; nothing here has yet been measured in this child's own cells.*

Sources report.md §3.1 / report.md §3.2 / methods-form.json Q14

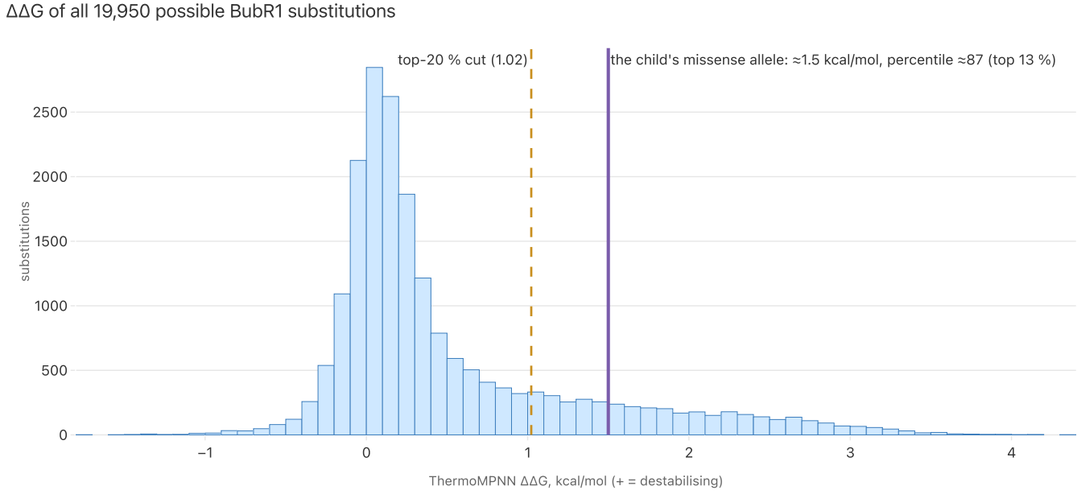
Does the typo make BubR1 unstable (55%) or inactive (45%)? Different answers point to different medicines.

+1.55 kcal/mol
ThermoMPNN ΔΔG — 87.3rd percentile of all 19,950 BubR1 swaps

55%
prior that mutant BubR1 is destabilised and degraded

45%
prior that it folds but sits on the job

AUROC 0.76
across 11 DMS proteins (0.60 on 8 MVA alleles) → this protein: **0.31–0.88**



All 19,950 possible BubR1 substitutions; the child's missense sits at +1.55 kcal/mol, the 87.3rd percentile of destabilisation. report.md §3.2 (entry 149); ThermoMPNN on AlphaFold v6, $0.015 of GPU

- **If unstable** the protein folds and is then thrown away — the job is to *keep* it, by shielding it from degradation.
- **If inactive** raising the amount of a dead enzyme buys little — the job is to clean up the senescent cells it leaves behind.

We did not hide the weak link — we tested it

In-gene calibration on n = 8 MVA alleles reached only AUROC 0.60. Across 18,825 buried core variants in 11 independent DMS proteins, ThermoMPNN reaches median AUROC 0.76 (95% CI 0.72–0.78), but for a new protein +1.55 kcal/mol predicts low abundance anywhere from 0.31 to 0.88. So 55/45 stays, and the blot decides.

One binary question sits upstream of every candidate drug.

Caveat *55/45 is an assumed weight. The 11-protein DMS calibration cannot replace it: its new-protein 95% interval (0.31–0.88) spans both branches.*

Sources report.md §3.2.2 / report.md §6.1 / report.md §11 (H083, H084)

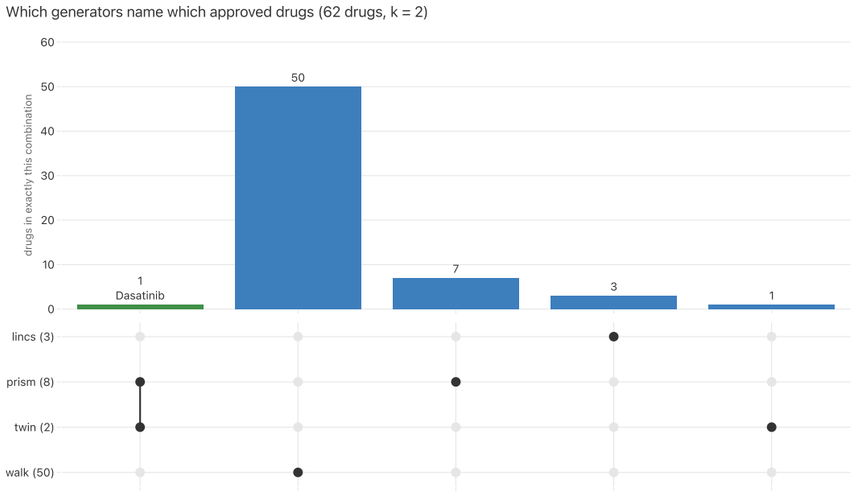
Every generator met a pre-registered null or control first — and we report the ones that failed.

21
hypotheses pre-registered before execution

8
came back refuted or uninterpretable — and are reported

p = 0.263
4-generator convergence is at chance (10,000-draw null)

0.33
readthrough context score — stop-skipping is deprioritised



Four independent generators; exactly one drug (dasatinib) is named by two of them, which a 10,000-draw null expects by chance (p = 0.263). report.md §5.1 (entry 155); figure step10-s2-upset

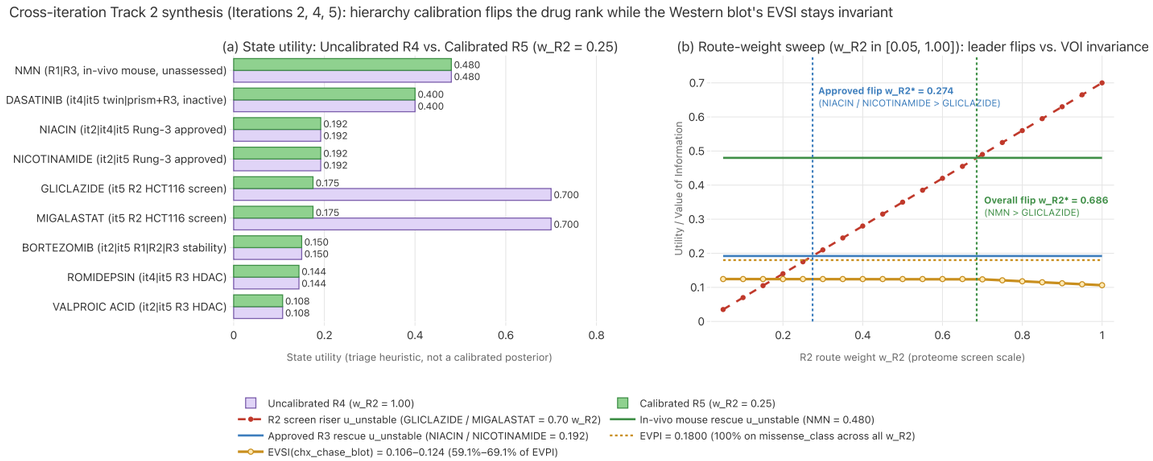
A naive pipeline would say	What our executed test found
Inhibit the checkpoint: Mps1/TTK, Aurora, PLK1	**DO-NOT** 33 of 34 are *less* effective in aneuploid lines (919 PRISM lines, p = 0.0096)
`stop_gained` → **give a readthrough agent**	**FUTILE** context score 0.33; EVPPI on readthrough = 0.0000
Reverse the knockout signature in LINCS	**CONFOUNDED** 11 of 12 hallmarks are generic arrest; 0 of 3,849 drugs pass FDR < 0.05
Weigh one cell-line screen like an animal study	**CALIBRATED** 20-step weight sweep restores the hierarchy at w_{R2} = 0.25

The refutations are the product: they set the weights everything else is scored with.

Caveat *Convergence on dasatinib is at chance level (p = 0.263) — it is a lead supported by two sources, not a proof.*

Sources report.md §5.1 / report.md §5.2 / report.md §5.3 / report.md §11

52 candidates, one scoring contract: a single-replicate cell screen no longer outranks a live-animal rescue.



Panel A: uncalibrated, the one-replicate proteomic hits lead; at w_{R2} = 0.25, NMN and dasatinib do. Panel B: the 20-step sweep, its two exact flip points (0.2743, 0.6857), value of information held flat. report.md §6.1–§6.2 (Entry 163); route_weight_sweep.tsv

In vivo rescue

NAD⁺/SIRT2 deacetylation of Lys668 restores BubR1 and doubles $Bub1b^{H/H}$ mouse lifespan (PMID:24825348)

→

Senolytic clearance

dasatinib: approved analogue of p16 clearance in the twin mouse, 99.4th pct aneuploidy selectivity (p = 0.0096)

→

Cell-line proteomics

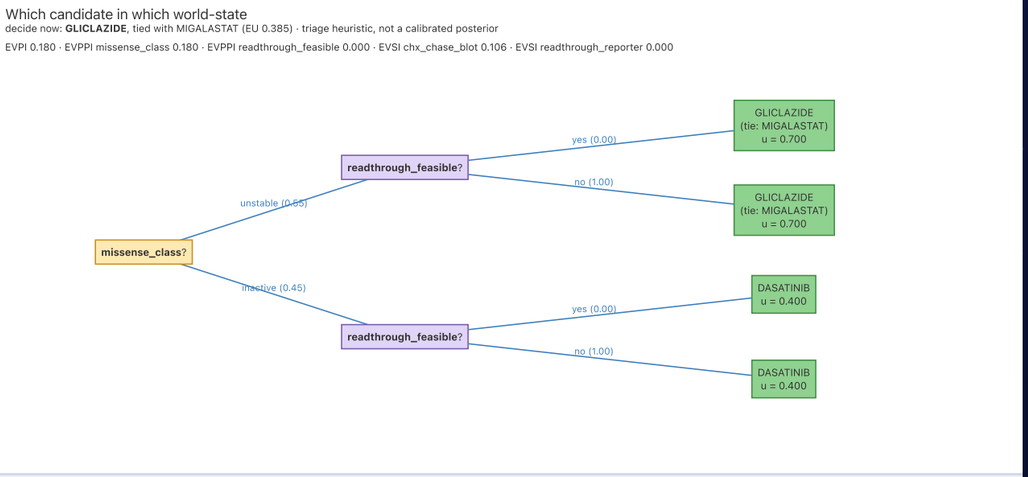
gliclazide, migalastat raise BubR1 2.7–2.8× among 875 perturbations — exploratory, discounted to w_{R2} = 0.25 after their own control failed

Ranking is a transparent, swept parameter — not a vibe.

Caveat *Route R2 is n = 1 in asynchronous HCT116 and its mitotic positive control failed (H099, p = 0.145).*

Sources report.md §5.2 / report.md §6.1 / report.md §6.2 / report.md §11 (H098, H099)

One blot in the child's own cells captures 69.1% nominal of the decision value — before anyone gives a first dose.



The tree as first built (uncalibrated R4): one assay, two branches. Calibrated R5 swaps the unstable-branch lead from gliclazide to NAD⁺ precursors. report.md §6.2 (entries 162, 163);...

Week 1–2 · measure — Cycloheximide-chase BubR1 Western blot, 0 / 2 / 4 / 8 h CHX ± MG132, patient vs parental fibroblasts or lymphoblastoid cells

↓

Rapid decay → P(unstable) = 0.874 — Branch A: NAD⁺ precursors — approved nicotinamide or niacin, NMN as the in-vivo-backed lead; migalastat / gliclazide exploratory

↓

Normal abundance → P(inactive) = 0.823 — Branch B: dasatinib — also the #1 approved lead if acting before the blot

↓

Either branch · tumour — simvastatin or lovastatin adjunct only after `SLCO1B1` and baseline CK

0.1800
EVPI — value of perfect knowledge, invariant across all 20 weights

100%
of that uncertainty sits on `missense_class`; 0% on readthrough

69.1%
of EVPI captured by the blot (EVSI = 0.1245) at Se 0.85 / Sp 0.90

The first measurement in the child's own cells

Every lead so far rests on the matched mouse or a cell line; zero candidates have been tested in this child. Branches are proposals for the treating team, not prescriptions.

We hand the tumour board a decision rule, not a ranked list.

Caveat *Utilities are model outputs under a refuted-calibration prior; zero candidates have been tested in this child.*

Sources report.md §0 row 2 / report.md §6.2 / report.md §8 step 2 / report.md §11 (H101, H102)

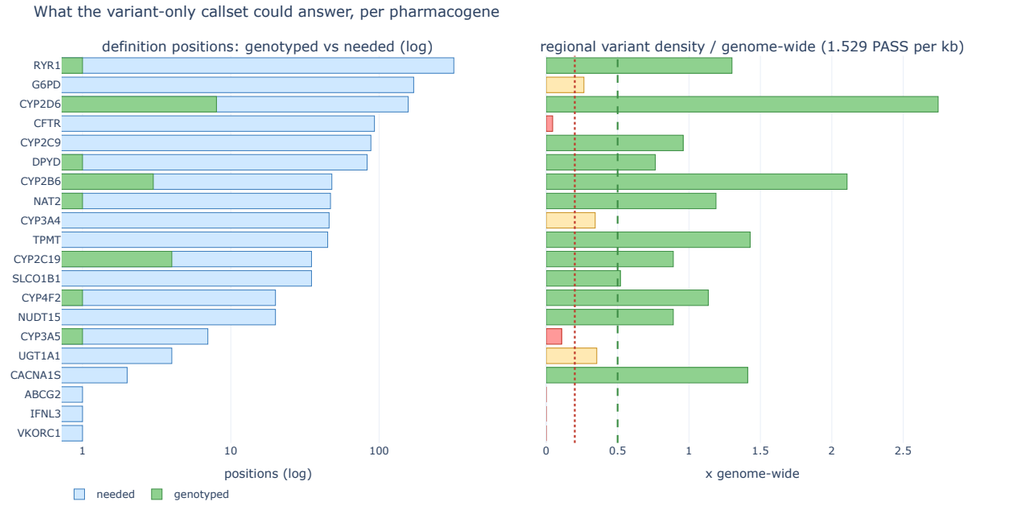
What *not* to give is a first-class output — audited against this child's own genome and symptoms.

33 / 34
Mps1/Aurora inhibitors are *less* effective in aneuploid lines (p = 0.0096)

1 drug
vincristine — the only candidate flagged by **both** safety sources

1.6%
of PharmCAT loci are genotyped by the delivered VCF (20 of 1,223)

5 genes
clinical PGx panel requested on day 1 before any statin



Only 20 of 1,223 pharmacogenomic positions are callable; G6PD sits in a low-callability zone. report.md §7.2 (entries 152, 161); figure step9-e56-coverage

- **DO-NOT** SAC-weakening inhibitors (Mps1/TTK, Aurora, PLK1, KIF11) — hard-excluded, reproducing PMID:33505028.
- **MONITOR** Vincristine stays an oncology decision; we raise a neuromuscular monitoring alert, not a discontinuation order.
- **GATE** Incidental `RYR1` VUS recorded as an anaesthesia precaution; statins gated on `SLCO1B1`.

We say "unflagged by PharmCAT and OnSIDES" — never "proven safe".

Caveat *A missing variant in a variant-only VCF is not evidence of wild type; that is exactly why we order the panel rather than assume.*

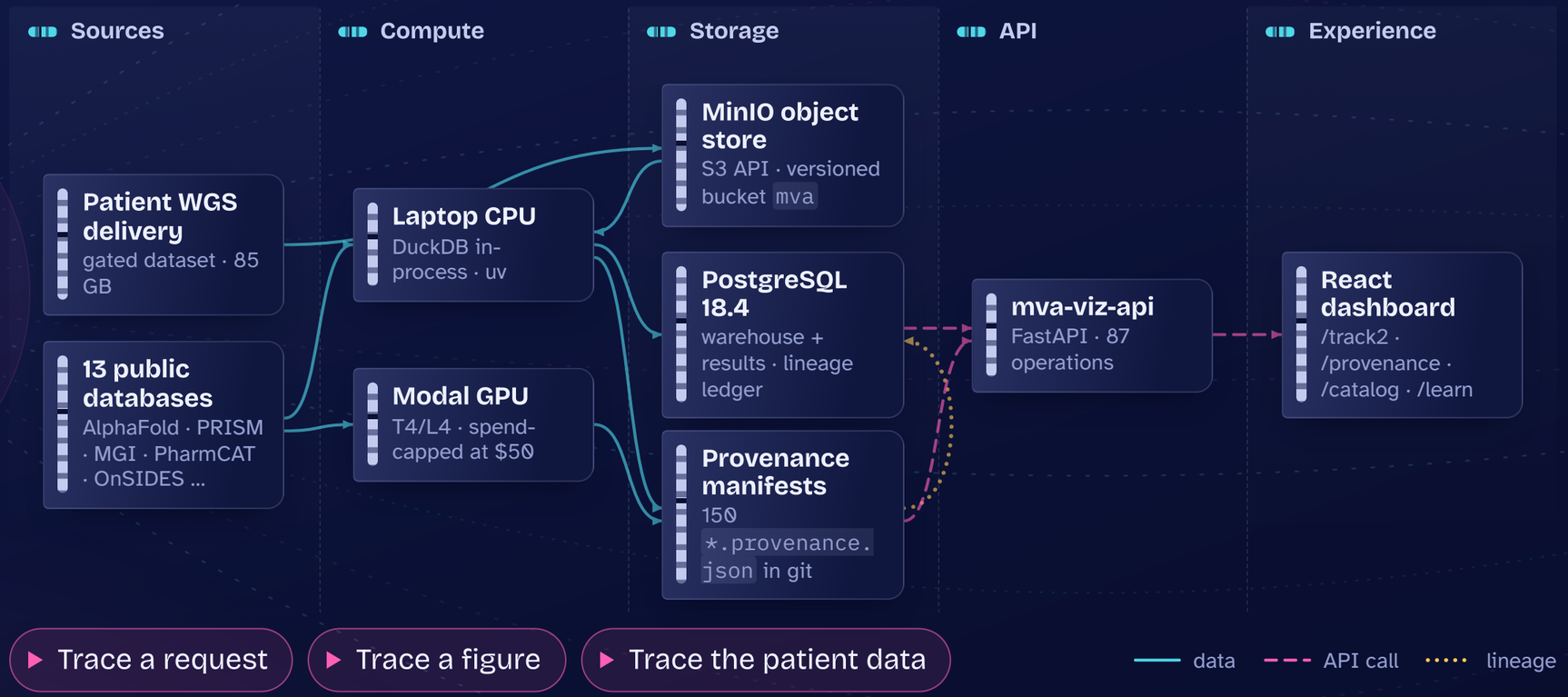
Sources report.md §7.1 / report.md §7.2 / report.md §7.3 / report.md §8

Every number you just saw comes out of one machine: object store, warehouse, hashed manifests, one API, one dashboard.

84.99 GB patient delivery in a versioned MinIO bucket — 12 / 12 objects SHA-256 verified

87 API operations (81 GET · 6 POST), one envelope `{data, cached_at, source}`

$0.16 Track 2's whole GPU bill — three small Modal jobs; everything else ran on a laptop CPU



From delivered bytes to the screen you are looking at

Sources	Patient WGS delivery, 13 public databases
Compute	Laptop CPU, Modal GPU
Storage	MinIO object store, PostgreSQL 18.4, Provenance manifests
API	mva-viz-api
Experience	React dashboard

7 data flows **3** API calls **1** lineage link

Select a component to see what it holds, or trace a request through the stack.

One storage layer, one API envelope, one dashboard — so any number can be followed back to where it came from.

Caveat *Research infrastructure, not production: one VPS with a single Postgres and MinIO. The Vercel-hosted dashboard is built and tested but not deployed; it runs locally or on committed fixtures.*

Sources docs/current-state.md §2 / docs/step-0/infrastructure.md §1–3 / src/mva/viz/app.py · response.py / report.md §10.2

Pick a headline claim: it walks back through its table, its script and commit, to SHA-256-hashed source files.

150 `*.provenance.json` manifests in git — inputs and outputs hashed

3,656 Python tests per run (+483 app tests), all hermetic — no patient data, no services

11 gates on every push, including one that fails the build if a patient coordinate leaks

Three headline claims, walked back to their bytes

- EVSI = 69.1% nominal of EVPI
- ΔΔG +1.55 kcal/mol · 87.3rd pct
- 33 of 34 Do-Not, p = 0.0096

source
decision_config.json + 17 more inputs
18 hashed inputs: the stability atlas, the three route tables, three…
`b4afb1b451ad…`

→

script
tools/it5_unified_synthesis.py @ fb9c708
Entry 163: sweeps w_{R2} from 0.05 to 1.00 in 20 steps; reproduces the…
no hash

→

artifact
route_weight_sweep.tsv
20 rows · 1,827 B · manifest built 2026-10-20
`76787b4e223b…`

→

artifact
row w_{R2} = 0.25
`evpi 0.18 · evsi_blot_chx 0.124455 ·…`
no hash

→

claim
69.1% of EVPI
Slide 5 KPI · report.md §6.2
no hash

5 steps **EVSI = 69.1% nominal of EVPI** traced from decision_config.json + 17 more inputs to the claim; 2 of 5 steps pinned by a SHA-256 content hash. Slide 5 KPI · report.md §6.2

Or ask the API yourself

`GET /api/provenance/lineage`

`GET /api/track2`

`GET /health`

every lineage edge, newest first, read live from Postgres


```
{
  "data": [{"target_schema": "track2",
"target_table": "prism_replication",
"source_kind": "file", "source_ref":
"https://ndownloader.figshare.com/files/46
630984", "source_sha256":
"3b6554cfc6c7…"}, …],
  "cached_at": null,
  "source": "pg.warehouse.data_lineage"
}
```


We do not ask you to trust a number: each one names its table, its script and commit, and the hashes of its inputs.

Caveat *Iteration 6's synthesis tables have manifests, but their ledger append is not yet recorded. Ledger and table counts date from Sept 2026 and were not re-counted live; on-disk hashes were re-checked 22 Oct.*

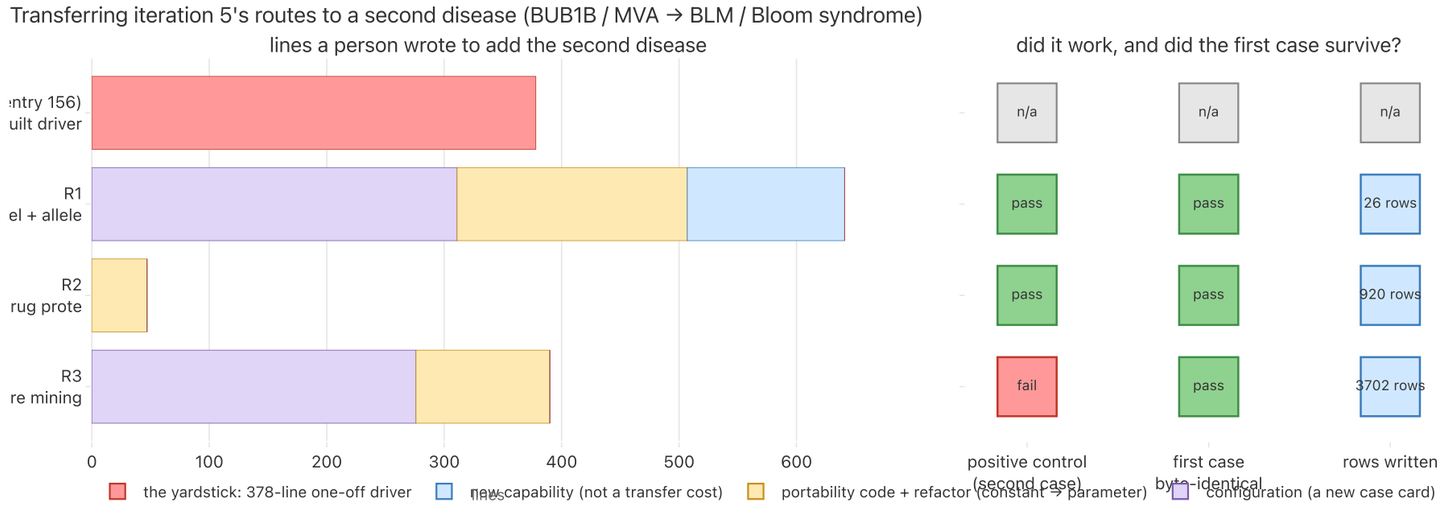
Sources ressources/track2/**/*.provenance.json / src/mva/warehouse/track2_lineage.py / src/mva/viz/response.py / tools/check-python.sh

Bloom syndrome ran end to end in 53 minutes for $0 — and we measured what adding a disease really costs.

53 min · $0
whole chain on Bloom syndrome (`BLM`), laptop CPU

top 1.15%
`Blm` mice recovered as the phenotype twin among 76,699 MGI genotypes

0.095
Jaccard vs the MVA list — disease-specific; cisplatin flagged Do-Not



Iteration-5 routes moved to Bloom: lines written (left; red = the 378-line Iteration-4 driver) and whether each passed its positive control and left the MVA case byte-identical (right). figure s7-case-card-transfer (Iteration 5); report.md §10.1

Zero-edit claim refuted (H096)
Iteration 4 needed a 378-line driver for 4 stages. That audit is why the Iteration 5–6 modules take the disease as parameters.

Re-tested on Bloom
R1 and R2 passed their positive control on the second case; R3 failed, and we report it. The MVA outputs stayed byte-identical.

Scalability claimed only as far as measured: a second disease is a case card plus counted edits, not a rewrite.

Caveat *Not zero-edit: H096 was refuted, and some code is still written per disease (line counts in the figure).*

Sources report.md §10.1 / report.md §10.2 / report.md §11 (H096) / figure s7-case-card-transfer

3:44 – 4:00 | Potential impact · 25%

Three things the treating team can start this week — and one experiment that decides the rest.

Day 1
Order the 5-gene PGx panel (`SLCO1B1`, `ABCG2`, `CYP2C9`, `G6PD`, `UGT1A1`); chart the vincristine and `RYR1` alerts

→

Week 1–2
Cycloheximide-chase BubR1 blot in the child's cells vs parental — 69.1% nominal of EVPI

→

Then
Low abundance → branch A (P = 0.874); normal → branch B (P = 0.823)

Everything is open and falsifiable

52 candidates, 13 public databases, 21 pre-registered hypotheses with 8 refutations kept, 150 SHA-256-hashed provenance manifests — all reproducible from tracked scripts.

Stated plainly

These are computational and translational hypotheses for research and clinical-team discussion. They are not clinical recommendations, and no candidate here has been tested in this child.

Order the panel, chart the alerts, run the blot — then choose the medicine with the evidence in hand.

Caveat *NMN and dasatinib evidence comes from the* $Bub1b^{H/H}$ *mouse; gliclazide and migalastat from one cell-line screen.*

Sources report.md §0 / report.md §1 / report.md §8 / README.md §1

Final audit: assumptions and portability

Sensitivity is conditional, not confidence

Joint grid: 3,815/7,425 settings (51.38%) capture ≥50% EVPI. EVPI .012–.240; blot EVSI 0–.2352. Canonical 69.1% nominal is not universal.

Third disease: partial real cached-public transfer

Fanconi/FANCA: 34 public HPO terms; Fanca rank 37/76,699 (symmetric), 102 (Phenodigm). Raw graph top-50 Jaccard .098901 vs MVA; .041667 vs Bloom. About 29s cached CPU; no live requests, no fixtures. No PRISM/LINCS, approval labelling, convergence, safety checks or therapeutic validation.

Evidence boundary

Eight allele abundance labels; stability AUROC .60 [.133,1] failed calibration. No measured patient prior or validated assay. Raw HCT116 10μM 24h BUB1B log2FC: gliclazide +.22, migalastat +.20; one summary column each. Earlier 1.47/1.44 are not raw BUB1B changes. No independent candidate replication or validated label expansion found.

Research triage only: priors, utilities and assay Se=Sp=0.85 are assumptions, not validated benefit or diagnostic probabilities. P(low)=0.535; P(normal)=0.465; P(unstable|low)=0.874; P(inactive|normal)=0.823. EVPI=0.18 is structurally prior(inactive)×0.4; zero readthrough information value uses strict dosage feasibility prior zero, not context heuristic 0.10. Primary patient/parental fibroblasts; CHX 0,2,4,8h ±MG132; culture and turnaround weeks 1–2, not a guaranteed weeks 1–2 clinical result. Embedded R4/R5 figures are historical model illustrations; any .90 specificity or .912/.831 posteriors in them are superseded by this .85-specificity model. No candidate is a prescription; approval metadata does not establish MVA efficacy.

Research triage only: priors, utilities and assay Se=Sp=0.85 are assumptions, not validated benefit or diagnostic probabilities. P(low)=0.535; P(normal)=0.465; P(unstable|low)=0.874; P(inactive|normal)=0.823. EVPI=0.18 is structurally prior(inactive)×0.4; zero readthrough information value uses strict dosage feasibility prior zero, not context heuristic 0.10. Primary patient/parental fibroblasts; CHX 0,2,4,8h ±MG132; culture and turnaround weeks 1–2, not a guaranteed weeks 1–2 clinical result. Embedded R4/R5 figures are historical model illustrations; any .90 specificity or .912/.831 posteriors in them are superseded by this .85-specificity model. No candidate is a prescription; approval metadata does not establish MVA efficacy. Historical R2 image labels 1.47/1.44 are not raw BUB1B log2FC; canonical raw values are .22/.20.